Robust News Retrieval Under Vocabulary Mismatch

A controlled comparison of lexical dense and hybrid retrieval

First Project Report

Project field	Details
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This proposal studies when exact lexical matching, semantic embedding retrieval, and their fusion succeed or fail when a user expresses the same information need with different words. The recommended design uses a standard news-search benchmark with existing relevance judgments, controlled meaning-preserving query rewrites, and a separate contrastive test for meaning-changing phenomena such as negation. This separation makes the results interpretable and avoids invalidating benchmark labels.

1 Project Objective

Primary objective: Determine how robust BM25, a modern dense bi-encoder, and a simple hybrid system are to vocabulary mismatch in news retrieval, and identify which observable query features explain their failures.

The project is not merely a demonstration that dense models understand synonyms or that BM25 rewards exact terms. It is a controlled robustness study. We hold the document collection and information need fixed, change only how the need is worded, measure the resulting performance change, and test whether hybrid retrieval reduces the weaknesses of either component.

1.1 Research Questions

1. RQ1 Baseline effectiveness: On original human-authored news topics, how do BM25, dense retrieval, and hybrid RRF compare?
2. RQ2 Vocabulary robustness: When a topic is rewritten without its most discriminative words but retains the same intent, how much does each system lose or gain?
3. RQ3 Failure modes: Which phenomena—paraphrase, acronym expansion, entity aliasing, temporal expression, and lexical specificity—predict the relative advantage of lexical or dense retrieval?
4. RQ4 Meaning sensitivity: On separately labeled contrastive examples, can the systems distinguish queries that differ in negation or exclusions?

5. RQ5 Complementarity: Does unsupervised rank fusion recover relevant documents missed by either single retriever?

1.2 Testable Hypotheses

- H1 Dense retrieval will show a smaller performance drop than BM25 on meaning-preserving paraphrases with low lexical overlap.
- H2 BM25 will remain competitive or superior for exact entities, rare names, numbers, and highly specific lexical constraints.
- H3 A rank-fusion system will improve average effectiveness and reduce worst-case query failures because lexical and dense rankings contain complementary relevant documents.
- H4 Neither vanilla BM25 nor a dense bi-encoder will reliably solve negation; a cross-encoder reranker will be a stronger reference system on the contrastive test.

2 Dataset Recommendation

2.1 Primary Benchmark

Recommended primary benchmark: TREC 2018 Common Core on the Washington Post Corpus version 2. It contains 50 ad-hoc search topics with title, description, and narrative fields, NIST relevance judgments, and approximately 608,000 news articles. This is a closer fit than MIND because the task is document retrieval for explicit information needs, not click-based news recommendation. It is also a closer fit than the TREC News background-linking task when the research question concerns short user queries.

Important access constraint: the Washington Post corpus is controlled-access. The organization must submit NIST agreements, and approval may take several business days. The topics and qrels are public, but the corpus is not an instant Hugging Face download. Access should therefore be requested immediately.

2.2 Role of TREC News

TREC News is a useful secondary transfer benchmark, not a direct substitute. Its background-linking task uses one news article as the query and asks for other articles that provide context. It tests semantic relatedness and temporal context in a realistic news setting, but it does not isolate short-query vocabulary mismatch. If time and corpus access permit, we will use its background-linking topics to test whether conclusions from ad-hoc search transfer to article-to-article retrieval.

2.3 Independent Negation Test

Use NevIR as a separate diagnostic benchmark. NevIR asks a retriever to rank two near-identical documents whose meaning differs by negation. Its labels and pairwise structure are appropriate for testing negation. These examples must not be mixed into the TREC aggregate score because they represent a different task and judgment scheme.

2.4 Dataset Decision Matrix

Dataset	Correct use	Strength	Risk or decision
TREC Common Core 2018 plus WaPo v2	Primary ad-hoc news retrieval benchmark	50 NIST topics and deep judgments on a real news corpus	Controlled access; request immediately

Dataset	Correct use	Strength	Risk or decision
TREC News background linking	Secondary transfer evaluation	Realistic article-to-article news context retrieval	Different task from short-query search
NevIR	Independent contrastive negation test	Purpose-built labels and clear pairwise metric	Not a full-corpus news search benchmark
MIND-small	Not recommended as primary	Large news metadata and recommendation logs	Click/impression labels do not answer ad-hoc relevance
Robust04	Fallback only if institution already holds it	Classic newswire ad-hoc benchmark	TREC disks are licensed; not instant-download
AG News	Pipeline smoke test only	Easy and small	Classification labels are not retrieval qrels

3 Proposed Experimental Pipeline

The pipeline deliberately separates benchmark evaluation, robustness evaluation, and contrastive diagnostics. This prevents a common design error: reusing old qrels after changing the meaning of a query.

Stage	Method	Output
1 Acquire and parse	WaPo v2 documents, Common Core topics, and qrels; preserve title, body, date, and entity strings	Canonical corpus and reproducible topic files
2 Build lexical index	Lucene BM25 through Pyserini; index title plus body; retain raw text and positions	BM25 ranked lists
3 Build dense index	Encode documents with E5-base-v2 or BGE-base-en-v1.5; L2-normalize; exact FAISS inner-product search initially	Dense ranked lists
4 Build hybrid baseline	Fuse BM25 and dense top-1000 ranks using reciprocal rank fusion with k=60	Hybrid ranked lists
5 Standard evaluation	Run all systems on the original 50 topics	nDCG@10, MAP, Recall@100 and per-topic scores
6 Create query variants	Two human-written, meaning-preserving rewrites per topic; remove or replace high-IDF anchor terms	100 paraphrase queries using unchanged qrels
7 Robustness analysis	Compare each rewrite with its source topic; group by rewrite phenomenon and lexical overlap	Delta metrics, confidence intervals, significance tests
8 Contrastive diagnostics	Evaluate NevIR and a small independently judged news exclusion set	Pairwise accuracy and qualitative errors
9 Optional reranking	Rerank the hybrid top 50 with a cross-encoder	Evidence on whether deeper interaction fixes failures

3.1 Query Variant Construction

For the primary robustness experiment, every rewrite must preserve the original information need so that the original qrels remain valid. Two group members independently verify each rewrite. Rewrites are tagged with one main phenomenon and checked against a short rubric.

Phenomenon	Allowed transformation	Example pattern
Synonym or paraphrase	Replace a content phrase while preserving intent	“job cuts” to “workforce reductions”
Acronym or alias	Switch between full name and common abbreviation	“International Monetary Fund” to “IMF”
Entity description	Replace a proper name with an unambiguous description	“the US central bank” for “Federal Reserve”
Temporal expression	Use an equivalent date or period expression	“during 2016” to “that year” only when context is explicit
Lexical de-anchoring	Remove a high-IDF term and restate the need	Replace a headline phrase with a natural description

Negation, number swaps, and entity substitutions are excluded from this set because they may change relevance. They belong in the separately judged contrastive evaluation.

4 System and Architecture Choices

4.1 Lexical Retriever

Use Pyserini/Lucene rather than rank_bm25. Lucene provides a standard inverted index, scalable retrieval, reproducible BM25 parameters, and direct production relevance. Start with k1=0.9 and b=0.4 as an explicit untuned setting, then tune only on a designated development subset or retain defaults to avoid test-set overfitting. Do not remove stopwords or stem by default; modern Lucene analysis and an ablation are safer than silently discarding negation terms such as “not”.

4.2 Dense Retriever

Use one retrieval-trained encoder as the principal dense baseline: intfloat/e5-base-v2 is recommended because it was trained for general text embeddings and uses explicit query and passage prefixes. BAAI/bge-base-en-v1.5 is a defensible alternative. all-MiniLM-L6-v2 may be retained only as a lightweight ablation because it is a general sentence-similarity model and is too weak to support broad claims about dense retrieval.

Encode each full document using a documented truncation strategy. A practical baseline concatenates title and the first 512 tokens. A stronger optional variant splits long articles into passages, retrieves passages, and aggregates by the maximum passage score. Report this choice because truncation can otherwise confound the lexical-versus-dense comparison.

4.3 Hybrid Retriever

Make reciprocal rank fusion a core baseline rather than a stretch goal. RRF combines ranks without requiring incomparable BM25 and cosine scores to be normalized. For document d, score(d) is the sum over

component rankings of 1/(k + rank(d)); use k=60 as the preregistered default and report any tuned alternative separately.

4.4 Storage and Database Architecture

No general-purpose database is required for the experiment. Use the data structure suited to each access pattern: Lucene for the inverted lexical index, FAISS IndexFlatIP for exact dense retrieval at this scale, and JSONL or SQLite for document metadata, query variants, annotations, and experiment manifests. This is simpler and more reproducible than adding Elasticsearch, PostgreSQL with pgvector, or a hosted vector database. If a deployment demonstration becomes necessary, OpenSearch can hold both BM25 and vector fields, but it should not complicate the scientific comparison.

5 Evaluation Plan

Question	Primary measure	Supporting analysis
Overall ranking quality	nDCG@10	MAP and Recall@100
Robustness to rewording	Mean delta nDCG@10 from original to rewrite	Worst-case drop and paired bootstrap 95% CI
Ranking stability	Per-topic score correlation and overlap of top results	Lexical-overlap bins and category breakdown
Negation sensitivity	Pairwise accuracy on NevIR	Error categories and cross-encoder comparison
System complementarity	Relevant-document overlap and oracle union recall	RRF gain over best single system

Statistical testing: use a paired randomization test or paired bootstrap over topics for system differences and report 95% confidence intervals. With only 50 topics, effect sizes and per-topic distributions matter more than a bare p-value. If several pairwise tests are run, apply Holm correction.

Annotation protocol: no new judgments are needed for meaning-preserving rewrites. For the small meaning-changing news set, pool the top 10 results from all systems, randomize and blind system identity, use a 0–2 relevance scale, double-label at least 20% of items, report Cohen's kappa or weighted kappa, and adjudicate disagreements.

5.1 Reproducibility Controls

- Freeze corpus version, model revision, tokenizer revision, query files, and package versions.
- Use deterministic seeds and save TREC-format run files for every condition.
- Separate development choices from final test reporting; never tune fusion or BM25 on the same topics used for the headline result without declaring it.
- Record latency, index size, and encoding time as secondary efficiency measures, using the same hardware.

6 Related Work and Its Role

Work	Finding used in this project	Design implication
BEIR, Thakur et al. 2021	BM25 is a robust zero-shot baseline; neural methods vary across domains	Use a strong standard lexical baseline and avoid assuming dense dominance
E5, Wang et al. 2022	Contrastive weak supervision produces transferable text embeddings	Use a retrieval-trained encoder with correct query and passage formatting
NevIR, Weller et al. 2024	Many neural retrievers perform near or below random on negation pairs	Evaluate negation separately and include a cross-encoder reference
RRF, Cormack et al. 2009	Simple rank fusion can outperform individual rankers without score calibration	Promote RRF to a core system
TREC Common Core 2018 overview	Provides deeply judged ad-hoc topics on the Washington Post corpus	Use existing qrels for a credible news-search evaluation
TREC News overview	Defines background linking as retrieval of useful context for a query article	Use only as a transfer task, not as short-query ad-hoc search

7 Key Innovations and Expected Contribution

- Paired query robustness: each original topic is compared with controlled meaning-preserving rewrites, so performance change can be attributed to wording rather than topic difficulty.
- Validity-preserving task separation: paraphrase robustness reuses qrels only when intent is unchanged, while negation and exclusions use independently valid labels.
- Failure prediction: query features such as lexical overlap, maximum IDF, entity density, numbers, and length are related to the BM25-versus-dense score difference.
- Complementarity analysis: the study measures not only final scores but also which relevant documents are uniquely recovered by each retriever and whether RRF reduces worst-case failures.

The contribution is empirical and methodological rather than a claim of a new retrieval model. A credible course-project outcome is a reproducible benchmark harness, a carefully constructed query-variant set, and evidence-backed recommendations for choosing lexical, dense, or hybrid retrieval under specific query conditions.

8 Stretch Objectives

- Cross-encoder reranking of the hybrid top 50 to test whether token-level query-document interaction improves negation and fine-grained constraints.
- Query-adaptive fusion: predict a BM25/dense fusion weight from pre-retrieval features, trained only on a development split and compared with fixed RRF.
- Long-document ablation: compare first-512-token encoding with passage-level dense retrieval and document aggregation.
- Transfer evaluation on TREC News background linking to test whether conclusions survive the change from short-query retrieval to article-context recommendation.

9 Milestones and Tentative Timeline

Milestone	Deliverable	Tentative timing
Access and setup	Submit WaPo request; acquire topics/qrels; reproduce a Pyserini sample run	Week 1
Baseline systems	BM25, E5 or BGE, FAISS index, TREC-format outputs	Week 2
Standard benchmark	Original-topic metrics, efficiency measurements, sanity checks	Week 3
Query variants	Two validated rewrites per topic with category labels	Week 3
Robustness study	Delta metrics, significance tests, overlap and feature analysis	Week 4
Hybrid and diagnostics	RRF baseline, NevIR evaluation, qualitative error analysis	Week 5
Stretch work and report	Reranker or adaptive fusion, figures, final report and demo	Week 6

10 Risks and Mitigations

Risk	Why it matters	Mitigation
WaPo access delay	Corpus approval can take several business days	Submit immediately; ask mentor whether institution already holds v2; begin code on a small open corpus only as a smoke test
Invalid reuse of qrels	Meaning-changing rewrites make old judgments wrong	Restrict main rewrites to intent preservation; judge contrastive items separately
Dense truncation bias	News articles exceed encoder input length	Document truncation; add passage retrieval ablation if time permits
Test-set tuning	Inflates apparent improvements	Pre-register defaults or create a development split; label tuned results clearly
Small sample uncertainty	Fifty topics limit statistical power	Use paired analysis, confidence intervals, effect sizes, and per-topic plots
Overambitious taxonomy	Too many categories leave few examples per group	Use five main rewrite categories and treat finer labels as qualitative

11 Immediate Decisions Needed from the Mentor

1. Can the group obtain the Washington Post Corpus v2 through the institution, or should the primary benchmark be changed before implementation?

2. Should the project remain ad-hoc search using Common Core topics, or should it become the TREC News background-linking task? These are related but distinct research problems.

3. Is the proposed scope—BM25, one strong dense retriever, RRF, controlled query variants, and NevIR—appropriate for the project duration?

4. May the group publish its query variants, labels, and code for reproducibility, subject to corpus licensing restrictions?

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